

Inferential Statistics: Knowledge Score vs. Residential Province

One-Way ANOVA & Tukey HSD Post-Hoc Analysis

This detailed report examines whether clinical knowledge differs significantly based on geographical residence, specifically comparing the Western Province, North Western Province, and all other provinces grouped together.

1. Descriptive Statistics

Province Group	N	Mean (Weighted V3)	Std Dev
Western	120	1.872	4.524
North Western	41	0.935	4.402
Other	40	3.337	4.636

2. Hypothesis Testing (One-Way ANOVA)

The overall test to see if geography matters.

- **F-Statistic:** 2.929
- **P-Value:** 0.0558
- **Conclusion:** Borderline/Not Statistically Significant ($p > 0.05$).

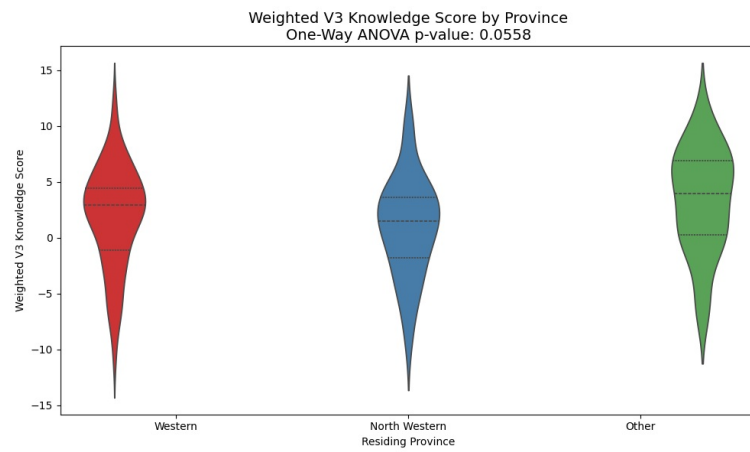
3. Post-Hoc Analysis (Tukey HSD)

Since the ANOVA p -value (0.0558) approaches significance, we run a pairwise Tukey HSD test to identify exactly which provinces differ.

	group1	group2	meandiff	p-adj	lower	upper	reject
	North Western	Other	2.4017	0.0466	0.0286	4.7749	True
	North Western	Western	0.9372	0.4872	-0.9945	2.8689	False
	Other	Western	-1.4646	0.1810	-3.4141	0.4850	False

Tukey Interpretation: If the reject column is True, those two specific provinces have statistically significantly different knowledge scores. If all are False, the variance between any specific pair is not large enough to declare significance independently.

4. Visualizations



Violin Plot